

SESSION CHECK

Session 4 — First light, then deep sky

Practical Astronomy Intensive

Name _____

Date _____

9 questions 30 minutes

Three of these nine are practical: setting a Dobsonian up, star-hopping to a Messier object with no GoTo, and handing in the log entry for it. The rest manage the expectations that wreck first nights — the Voyager postcard, the Hubble photograph, and the filter that is supposed to brighten a galaxy.

Circle one letter for each multiple-choice question. Write short answers on the lines. Tasks marked **Practical** are done, not written — your assessor will watch.

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- 1 A pair of binoculars is marked 7×50. What do the two numbers mean, and what should you expect of them?
 - a 50× magnification in a 7-element optical design — a modest toy to be outgrown once a real telescope arrives
 - b 7× magnification and 50 mm objectives, and since 7× is low they can be pushed handheld to 15× or 20× with no loss of steadiness
 - c 7× magnification and 50 mm objectives — useful for scenery, but they show nothing in the sky a telescope will not show better
 - d 7× magnification and 50 mm objectives — the 50 decides how faint you can go, and they will show Jupiter's moons, the Pleiades and hundreds of clusters in a field no telescope can match
 - 2 A student arrives at his first dark-sky night having taken his new Dobsonian out of the box that afternoon. He wants to start on M57. What is the order of business?
 - a Fit the highest-power eyepiece in the box first, since M57 is small and needs the magnification to be found
 - b Finder aligned in daylight, tube out early to cool, lowest-power eyepiece in, centre on something bright, then creep the focuser slowly — and only then go hunting
 - c Point it at Lyra and look — a telescope shows you things; that is what it is for
 - d Look through it once, and if the view is fuzzy return the scope — a fuzzy view means the optics are bad
 - 3 Set up the Dobsonian from its case and get first light on a bright star of your choosing. Work unaided; narrate what you are doing.

Practical — performed for your assessor.
 - 4 Using the chart and the finder only, star-hop to a Messier object the assessor names. No GoTo, no app, no help.

Practical — performed for your assessor.

- 5 Log the object you have just found, and sketch it. Hand the entry to the assessor before you move the telescope.

Practical — performed for your assessor.

- 6 You split a pretty double in the eyepiece. A student says the two stars must be orbiting one another. Is that safe to assume?
- a No — many are optical doubles: two stars at wildly different distances that merely happen to line up from where we sit
 - b Yes, and any double that will not split just needs more magnification to reveal the pair
 - c It cannot be settled by eye at all — questions like this, and variable-star work, need instruments rather than observers
 - d Yes — if two stars appear that close together in the eyepiece they must be a bound pair
- 7 Saturn is centred at 120× on a night of mediocre seeing. What gets you the most detail?
- a Take one good look, note it in the log, and move on to the next target while the sky is dark
 - b Expect the Voyager view at 120× and, if it is not there, conclude the scope is under-powered
 - c Stay where you are and keep watching — planetary detail surfaces in brief moments of steady air, and you have to be at the eyepiece when they come
 - d Reach straight for 300× — Saturn is small and bright, so more power is always the way to more detail on it
- 8 A student who has read the Messier table finds M31, looks, and says flatly: 'That's it? A grey smudge.' What is the honest answer?
- a Nothing is wrong with the view — it is the same at any site, so this is simply what M31 looks like everywhere
 - b Turn the power up until the smudge resolves into the photograph's detail
 - c The telescope is too small — a larger instrument would show the spiral arms and colour of the photograph
 - d Yes — visually these are faint grey smudges, because a camera integrates photons for hours and the eye cannot integrate over time at all; the skill is learning to see what IS there
- 9 An OIII filter makes the Veil Nebula appear out of nothing. A student immediately screws it in to look at M31. What happens, and why?
- a M31 improves enough that the filter is a fair substitute for driving out to a dark site
 - b M31 shows more contrast — the filter works on any faint fuzzy, galaxies included
 - c M31 gets brighter, as it did for the Veil — that is what a nebula filter does
 - d M31 gets dimmer and no easier — a filter subtracts light, passing only the narrow lines an emission nebula radiates, and a galaxy's broad starlight is blocked along with the skyglow